

Autonomous Tree Planting Robot to Combat Climate Change caused by Deforestation

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Project Origin

- Travelled to Punjab, India to my grandparents farm
- Saw that many of the crops were dead and the ground was extremely dry and cracked
- Found out the main cause of this, in the area was due to deforestation
- Due to the deforestation, the temperature of the area had significantly risen
- Crops were affected from this as many of them started drying up and dying
- Efforts have been made towards reforestation in Punjab but progress is not being made fast enough

Not a continuation project.

Work By Others

- Official Raspberry Pi website significantly contributed in helping me understand the wiring of the board. Along with this, large amount of documentation was given on how to control motors and servos
- The grooves made on the metal for the claw were also ordered. As this requires precision and a professional machine, these cuts had to be ordered



What and Why is it a Problem?

- Deforestation contributes 12-20% of global greenhouse emissions
- Last year 15 billion trees were cut down
- Only 1 billion trees were planted
- 44% of saplings planted last year, died due to not being planted properly
- 85% of trees planted in the United States were planted manually by volunteers

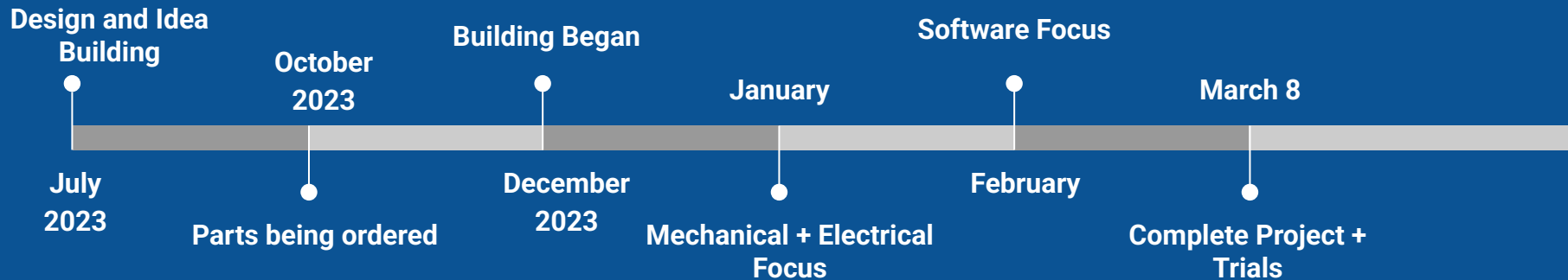
Experts have started to claim that we are approaching a point of no turn in the battle against global warming. Much of this is because our inability to solve deforestation and plant more trees. They have also stated that if we do not start making progress with reforestation, we could fall significantly behind in the race against the rate of deforestation.

Volunteer organizations have sprung up around the United States but many of them are not making a large effect. Companies have also started to make efforts to become Carbon Neutral and fund these volunteer associations. However, the effort is happening too slowly.

Drones and Robots are starting to be utilised in order to plant trees as well as seeds but development of these methods have not been well developed

Solution Overview & Timeline

- Robot that is able to autonomously plant trees
- Carries multiple saplings in plantable biodegradable containers
- Able to detect obstacles and maneuver past them
- Determine in which area to plant saplings based on given commands
- Detect rough terrain and determine whether to find an alternative route or not
- Plant continuously without assistance
- Aluminum Body to withstand any major bumps or damages



Procedure

The structure of the robot was built with large and medium sized C-Channels. Along with this, 4 plates were put in the middle of the robot to provide a base for the electronic components. With a strong base electronic components were started to be put together. The Raspberry Pi was put towards the back of the robot and the motor drivers were added. These were able to successfully drive wheels of the robot. Each pair of wheels was also connected to their own power source in order to ensure they move fast enough. Next the claw and arm was put together. These were put together through C-Channels and custom cut metal parts. Through this, the shape for the claw was built and it was able to grip the plant. These were connected to their own Servo Driver and also connected to their own power source. Through this, each of the six servos would be able to receive enough power. Finally, each of these components was programmed to move smoothly and as fast as possible.

Solution: Mechanical

Main Components

- 4 Omnidirectional Wheels with 4 DC motors
- 1 Large Claw
- Multidirectional Arm
- 1 Raspberry Pi 4B
- 2 DC Motor Drivers
- 1 Multi-Channel Servo Motor Driver
- 5 Servo Motors
- And many wires :-)

Arm

- Multi-Directional Arm with 5 servo motors
- Responsible for moving the claw and assisting in planting
- Connected to the servo driver
- Located at the back of the robot

Claw

- Operating using 1 servo motor
- Picks up and plants the sapling
- Able to rotate 360 degrees
- 180 degrees opening range

Movement

- Movement for the robot is conducted through omnidirectional wheels
- Two wheels connected to one motor drive
- Wheels made for going over rough terrain

Solution: Electrical

Main Components

- Raspberry Pi 4B with 17 General Purpose Input Output (GPIO) pins utilised
- 2 L298N Motor Drivers
- 1 PCA9685 16 Channel Servo Driver
- 2 6V Battery Packs with 2000 mAH current
- 1 5V Battery Pack with 120000 mAH current

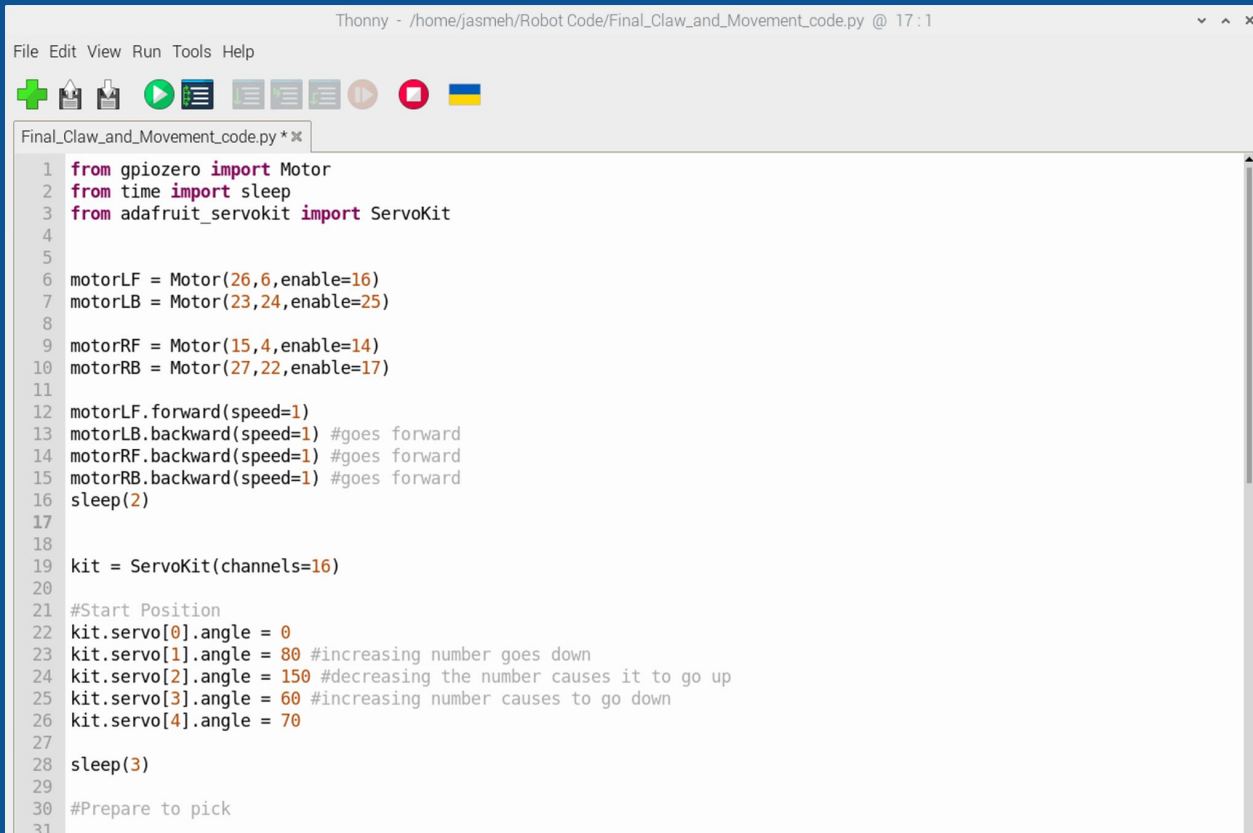
Motors Drivers

- 2 motor drivers responsible for controlling 4 DC motors
- Powered by 2 6V Battery Packs with 2000 mAH current
- Responsible for DC motor movement
- Connect to the Raspberry Pi and Battery Packs
- Control Specific turn angles and speed

Servo Driver

- Responsible for controlling 5 servo motors using Pulse Width Modulation (PWM)
- Powered 1 5V Battery Pack with 120000 mAH current
- Connected to Raspberry Pi
- 6 V Battery Connected
- Each Servo has 3 ports on the driver

Solution: Software



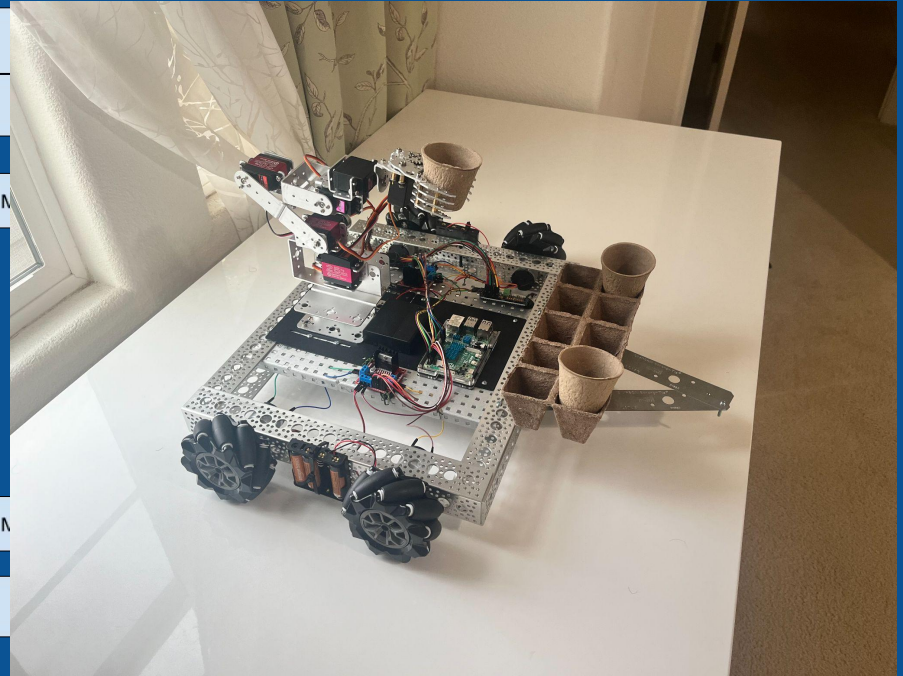
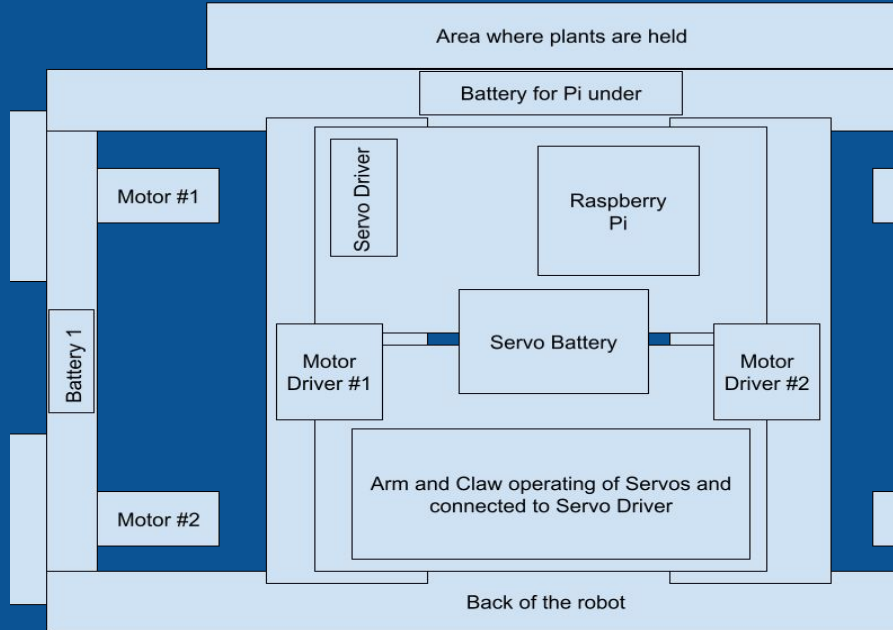
The screenshot shows the Thonny IDE interface. The title bar indicates the file path: Thonny - /home/jasmeh/Robot Code/Final_Claw_and_Movement_code.py @ 17:1. The menu bar includes File, Edit, View, Run, Tools, and Help. The toolbar contains icons for opening a file, saving, running, and other IDE functions. The main editor window displays the following Python code:

```
1 from gpiozero import Motor
2 from time import sleep
3 from adafruit_servokit import ServoKit
4
5
6 motorLF = Motor(26,6,enable=16)
7 motorLB = Motor(23,24,enable=25)
8
9 motorRF = Motor(15,4,enable=14)
10 motorRB = Motor(27,22,enable=17)
11
12 motorLF.forward(speed=1)
13 motorLB.backward(speed=1) #goes forward
14 motorRF.backward(speed=1) #goes forward
15 motorRB.backward(speed=1) #goes forward
16 sleep(2)
17
18
19 kit = ServoKit(channels=16)
20
21 #Start Position
22 kit.servo[0].angle = 0
23 kit.servo[1].angle = 80 #increasing number goes down
24 kit.servo[2].angle = 150 #decreasing the number causes it to go up
25 kit.servo[3].angle = 60 #increasing number causes to go down
26 kit.servo[4].angle = 70
27
28 sleep(3)
29
30 #Prepare to pick
31
```

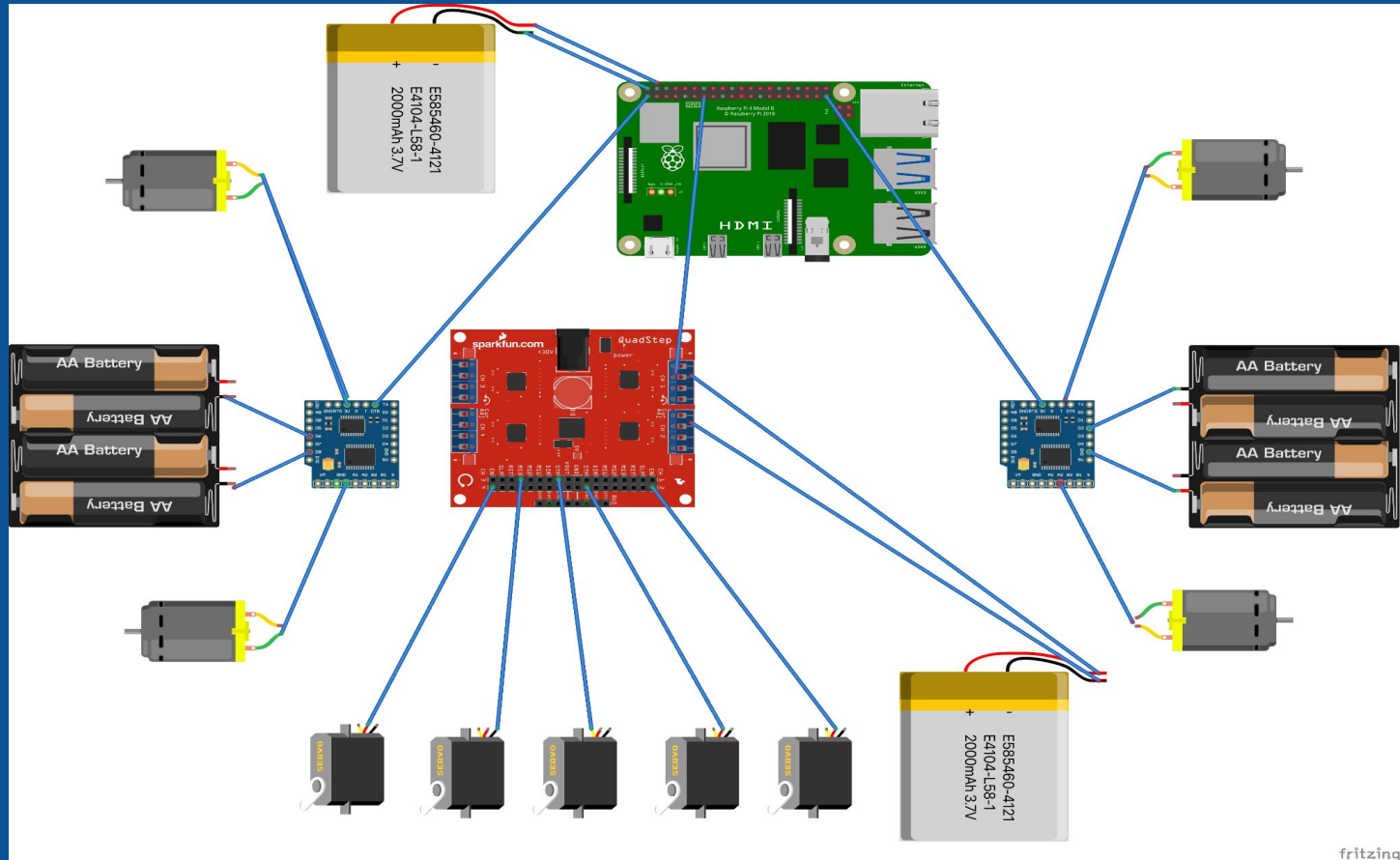
Overview

- Using Raspberry Pi OS
- Code written in Python 3 using Thonny IDE
- Use gpiozero library to control motor drivers and wheel motors
- Use adafruit-circuitpython-servo kit library to control servo motors, arm and claw

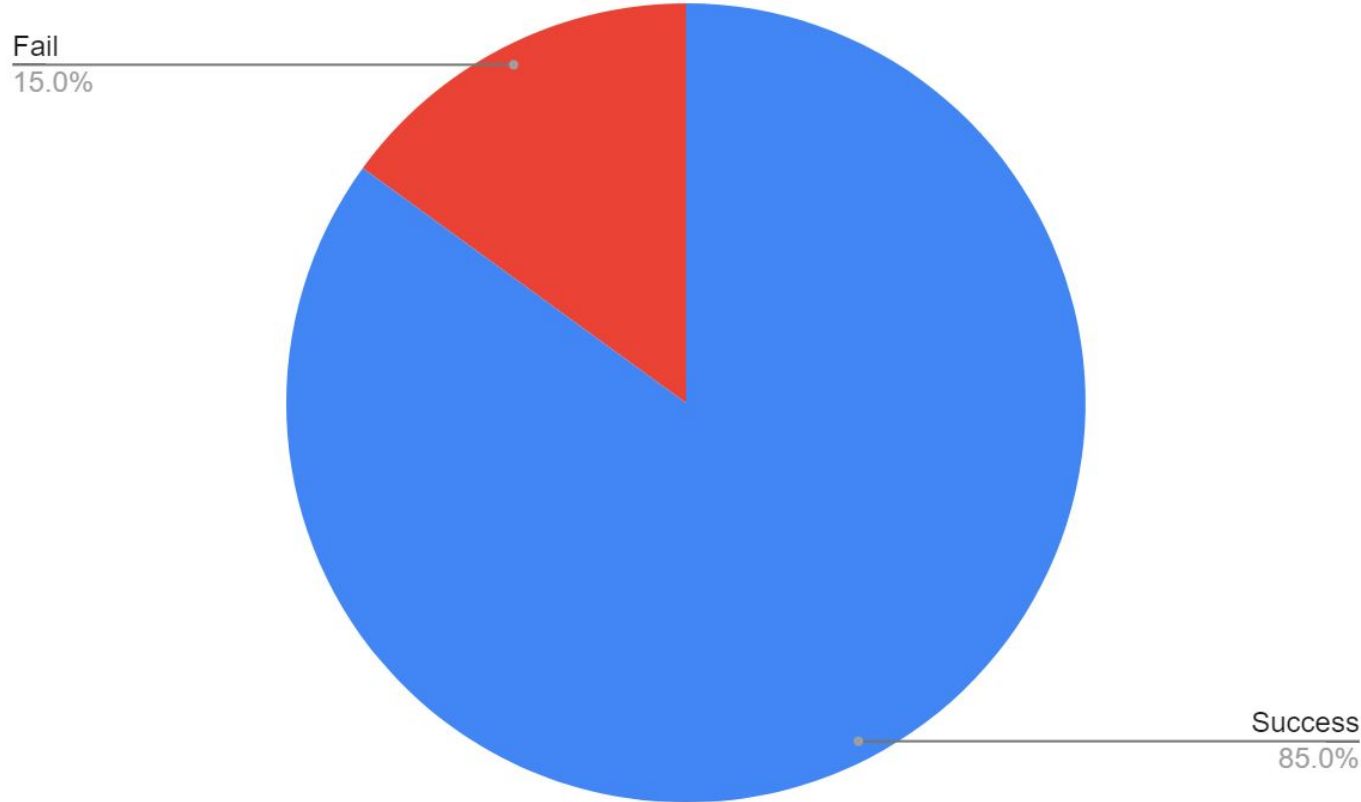
Design Schematic



Circuit Schematic



Trial Results of Pick and Plant



Results

- 20 trials conducted
- 17 successful trials of picking and planting the sapling
- 3 failed trials due to the robotic arm not being able to successfully plant the sapling

Learnings and Next Steps

Learnings

- Learned about the different GPIO pins on the Raspberry Pi and what function each serves
- Purpose of the DC Motor Drivers
- Purpose of Servo Driver
- Importance of using the right voltage / current to power different hardware
- Different Python programming libraries that seamlessly enable hardware-software communication
- No shortcuts...use and tighten all screws :-)

Next Steps

- Code is incomplete, each step of the planting process is hard coded
- Computer vision should be added to enable the robot arm to detect where the saplings is and automatically pick it up
- Geo-spatial awareness to determine area of land that needs to be planted
- A better system to dig holes has to be added to the robot
- More space to carry more saplings
- More sensors in order to receive input from behind the robot as well

References

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